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A multi-species comparison of selective placement patterns of ramets in invasive alien and native clonal plants to light, soil nutrient and water heterogeneity

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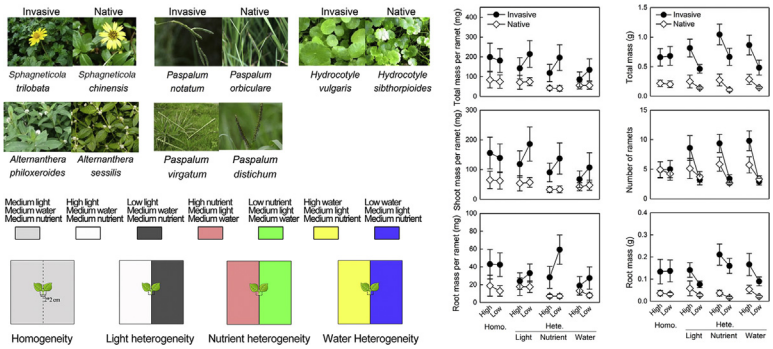
HIGHLIGHTS

- Selective replacement of ramets in clonal plants can maximum utilize resource and share risk in heterogeneous environments.
- Invasives benefit more from resource-high and resource-low patches than natives in all heterogeneous environments.
- Selective placement patterns promote higher morphological plasticity and performance for invasives than natives.
- Selective placement patterns of ramets might play important role in alien clonal plant performance and competitive superior.

GRAPHICAL ABSTRACT



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ABSTRACT

A worth noticing pattern in current invasive biology is the clonal ability of many of the world's worst invasive plants. Selective placement of ramets (i.e. foraging behavior) can intensify ramet performance and allocation, and place more ramets in the more favorable microhabitats, which can maximum utilize resource and share risk in heterogeneous environments. Still little is known about whether invasive alien and native clonal plants differ in the selective placement patterns of ramets in invasive clonal plants or not.

We used five congeneric pairs of naturally co-occurring invasive alien and native clonal plant species in China. In a glasshouse, we grew all species in pots under a homogeneous and three heterogeneous conditions (i.e. light, soil nutrients or water) subjected to resource-high or -low patches.

All biomass parameters and number of ramets significantly increased in resource-high patches in all three types of heterogeneous environments. Interestingly, growth of invasive alien plants benefited significantly more from resource-high patches than native plants in all heterogeneous environments. Overall, invasive had higher biomass parameters per ramet than natives. Ramet parameters of invasive plants also benefited more from resource-low patches than natives. Three different selective placement patterns of ramets in resource-low patches were exhibited in invasive plants: ramet increasing shoot investment (above pattern), increasing root investment (below pattern) and increasing both investments (complete pattern) in the light, soil water and

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nutrient heterogeneity, respectively. Investment on less, larger ramet was the adaptive strategy of invasive plants in resource-poor patches.
The results suggest that adaptively selective placement patterns of ramets promote a higher morphology plasticity and performance in invasive clonal plants over natives. When alien clonal plants spread new areas with light, soil nutrients or water heterogeneity, selective placement patterns of ramets might play an important role in plant performance and competitive superior by capitalizing more on additional resources.
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1. Introduction

Biological invasion has threatened biodiversity, economic development and ecosystem functions globally (Callaway and Aschehoug, 2000; Rejmánek, 2015; Richardson et al., 2000; Van Kleunen et al., 2010, 2015). Competitive interactions between the alien and native species are considered important for successful invasion (Davidson et al., 2011; Funk and Vitousek, 2007; Pearse and Altermatt, 2013; Qin et al., 2013; Sun et al., 2015; Whitney and Gabler, 2008). The biological characteristics of invasive alien plants have a direct impact on their invasiveness (Trtikova, 2009) and a worth noticing pattern in current invasive biology is the clonal ability of many of the world's worst invasive and other invasive alien plants (Liu et al., 2006; Lowe et al., 2000; Pyšek, 1997; Song et al., 2013; Yu et al., 2009). Such arguments raises questions whether clonal traits exist in the invasive clonal plant species that confers them an advantage over native clonal or non-clonal plants.

Phenotypic plasticity, as the change expressed in the morphology performance of a genotype under different environments, is often mentioned as a key trait that promotes and enhances the invasiveness of alien plants (Baker, 1965; Hulme, 2008; Richards et al., 2006; Van Kleunen and Fischer, 2001). Some studies have shown higher plasticity in invasive species than in native species (Davidson et al., 2011; Drenovsky et al., 2008). Selective placement of ramets (i.e. foraging behavior) is an important plastic trait for resource uptake in clonal plants. It can intensify ramet performance and allocation, and place more ramets in the more favorable microhabitats/patches via sharing and translocating of resources such as water, carbohydrates and mineral nutrients in the same clone (Caraco and Kelly, 1991; De Kroon and Hutchings, 1995; De Kroon et al., 1998; Van Kleunen and Fischer, 2001). Thus selective placement of ramets in clonal plants can maximum utilize resource and share risk within genet, and cause differential morphology plasticity of plants in natural environments. Studies have shown that root-foraging can greatly increase performance of invasive alien plant species (Chen et al., 2017; Keser et al., 2014, 2015). A meta-analysis was only focused on morphological foraging traits (i.e. spacer length, specific spacer length, branch intensity and branch angle) of clonal plants in light, nutrients and water resource (Xie et al., 2014). However, for many of the world's worst invasive clonal plants, still little is known about whether invasive alien and native clonal plants differ in their patterns of selective placement of ramets and whether these differences will promote a higher morphology plasticity and performance in invasive clonal plants or not.

Environments often varied spatio-temporally, and key resources such as light, soil nutrients and water are generally spatially distributed in patterns of heterogeneity even at a centimeter scale (Gascha et al., 2015; Jackson and Caldwell, 1993; Peipoch et al., 2016). Individuals of many clonal plants can spread large areas by fast clonal growth, and they are likely to experience light, soil nutrient or water heterogeneity within the reach of their clonal growth organs (e.g., rhizomes and stolons) or root systems. They can potentially modify the morphology of their clonal growth organs and roots (De Kroon and Hutchings, 1995; Semchenko et al., 2007) in response to resource-rich and resource-poor patches. Selective placement of ramets is the one of most significant trait that confer clonal plants with an advantage to enable plants to better use the heterogeneous environments (Alpert, 1999; De Kroon and Hutchings, 1995; Keser et al., 2014, 2015; Wang et al., 2008; Semchenko et al., 2007).

In the glasshouse experiments, isolated ramet of five congeneric and co-existing invasive and native clonal plant species were planted in a homogeneous treatment and three treatments that heterogeneous distributions of light, soil water or nutrients. We simulated environmental heterogeneity by creating patches of low (poor) and high (rich) levels of these three resources in a pot. We hypothesize that in resource-rich patches, invader might invest more root, clonal organs and ramets to size superiority over natives, and in resource-poor ones, invader might increase the uptake ability of resource to overcome natives. Then this integrative cooperation will promote plasticity performance of the whole plant.

2. Materials and methods

2.1. Species selection and cultivation

We chose five pairs of clonal plant species in which three pairs are stoloniferous clonal plants and the other two pairs are rhizomatous ones (Table 1). Two species in each pair are congeneric and coexisting invasive alien and native in China, respectively, which avoid largely differ in their habitat preferences and phylogenetic relatedness (Felsenstein, 1985). The 10 species are from four genera (*Wedelia*, *Alternanthera*, *Hydrocotyle* and *Paspalum*) of four families (Asteraceae, Amaranthaceae, Araliaceae and Poaceae; Table 1).

All plants were collected in Zhejiang, Guangdong and Hubei Province. We collected mixed populations including >100 ramets from five locations a distance of at least 500 m apart to increase the similarity to sample ramets from different genets (i.e. genotypes) for each species. All ramets were collected in 2014, and vegetatively cultivated for about six months in a glasshouse at Huazhong Agricultural University in Wuhan, Hubei Province, China. In 2014, we selected 32 similar-sized ramet for each species, and used them for the experiment described below. Each ramet had three leaves and few roots.

2.2. Experimental design

The study was conducted in a plastic glasshouse at Huazhong Agricultural University, Wuhan, China. On 19 August 2014, a square pot (24 cm long × 24 cm wide × 20 cm deep) separated into two equal parts (24 cm long × 12 cm wide × 20 cm deep) with a plastic divider was used and each ramet was planted in the center of pot. The divider was stuck to the two side-walls and bottom of the pot, and a small vertical opening of 2 cm × 2 cm was created at the upper, central part of the divider to allow the planting of root of the ramet (Fig. 1). After the ramet was planted, the opening in the divider was sealed with sea mud in space between root and opening boundary to prevent movement of water and nutrients between the two compartments of a pot. Then we covered the upper of divider with experimental soil in the pot. The pots were filled with 7.2 L mixture of fine river sand and yellow-brown soil (v:v, 1:1) collected from Shizishan Mountain in Wuhan, Hubei Province, China. The nutrient concentration of the soil mixture was quite low, with a total N of 0.58 ± 0.06 g kg^{-1}, a total P of 0.32 ± 0.04 g kg^{-1} and a total K of 18.96 ± 1.01 g kg^{-1} (mean ± SE, n = 8).

Table 1
Clonal plant species used in the experiment.

Species	Family	Origin	Native range	Clonal organ	Typical habitat
Sphagneticola trilobata (L.) Pruski.	Asteraceae	Invasive alien	North and South America	Stolon	Moist grasslands, edges of canals, roadsides
Sphagneticola chinensis (Osbeck.) Merrill	Asteraceae	Native	Asia	Stolon	Moist grasslands, edges of canals, crop fields, roadsides
Alternanthera philoxeroides (Mart.) Griseb	Amaranthaceae	Invasive alien	South America	Stolon	Wetlands, canals, nearby fields
Alternanthera sessilis (Linn.) DC	Amaranthaceae	Native	Asia, Africa	Stolon	Wetlands, other moist habitats
Hydrocotyle vulgaris L.	Araliaceae	Invasive alien	North America, Europe	Stolon	Wetlands, other moist habitats
Hydrocotyle sibthorpioides Lam.	Araliaceae	Native	Asia	Stolon	Wetlands, other moist habitats
Paspalum notatum Flugge	Poaceae	Invasive alien	North and South America	Rhizome	Roadsides and grasslands
Paspalum orbiculare Forst.	Poaceae	Native	Asia, Oceania	Rhizome	Roadsides, other moist habitats
Paspalum virgatum L.	Poaceae	Invasive alien	South America	Rhizome	Moist grasslands
Paspalum distichum L.	Poaceae	Native	Tropics and subtropics of Asia, America	Rhizome	Roadsides, nearby fields, grasslands

Origin and habitat information are based on the Flora of China (www.efloras.org), Scientific Database of China Plant Species (DCP) (http://www.plants.csdb.cn/eflora) and other reference Wang et al. (2017).

After allowing the plants to recover for two weeks (on 3 September 2014), we started the experiments to test effects of origin (alien vs. native species) and resource heterogeneity (homogeneity vs. heterogeneity, i.e. homogeneity vs. light heterogeneity, homogeneity vs. nutrient heterogeneity and homogeneity vs. water heterogeneity, respectively) in a homogeneous distribution and three heterogeneous distributions of light, soil water or nutrients. Four treatments (i.e. the pot in each treatment was divided into resource-high and resource-low patches) are as follows (Fig. 1): (i) homogeneity (actually both resource-high and resource-low patches received the same medium light, medium water and medium nutrient), (ii) light heterogeneity (one received high light with medium water and medium nutrient, and the other received low light with medium water and medium nutrient), (iii) nutrient heterogeneity (one received high nutrient with medium light and medium water, and the other received low nutrient with medium light and medium water) and (iv) water heterogeneity (one received high water with medium light and medium nutrient, and the other received low water with medium light and medium nutrient). The light, water and nutrient received by the plants in the heterogeneous treatments was the same as that in the homogeneous treatment. We defined and marked the resource-high and resource-low part of a pot at the first time. Then we randomly changed the direction and position in glasshouse weekly.

High light was 100% of natural light in the glasshouse, and low light was about 10% of high light by covering the resource-high and resource-low patches of the pot with two layers of black shading net, respectively. We created medium light (i.e. 53–57% of natural light in the greenhouse) by covering the patches with one layer of shading nets. A cuboid shading structure (24 cm long × 12 cm wide × 40 cm high) that surface covered with different black shading net was set up in the above three light patches. High soil nutrients were conducted by adding a nutrient solution containing 0.2 g water-soluble fertilizer (20%–20%–20%, N-P_2O_5-K_2O; Peters Professional, Scotts, Geldermalsen, The Netherlands) to the resource-high patch once per week, and low nutrients were conducted by adding the same volume of water (no fertilizer) to the resource-low patch of the pot. Medium soil nutrients were conducted by adding a nutrient solution containing 0.2 g water-soluble fertilizer to the whole pot once per week. High soil water availability was created by supplying the resource-high patch with 200 mL water every one to four days depending on weather conditions and how fast the soil dried out. Low water was created by adding only 20 mL to the part with the resource-high patch of the pot and medium water was created by adding only 220 mL to the whole pot at the same time. Soil water content was 35.23 ± 3.68% (mean ± SE, n = 8) in the high water, 25.03 ± 2.94% (mean ± SE, n = 8) in the medium water and 12.58 ± 2.63% (mean ± SE, n = 8) in the low water, as measured every day

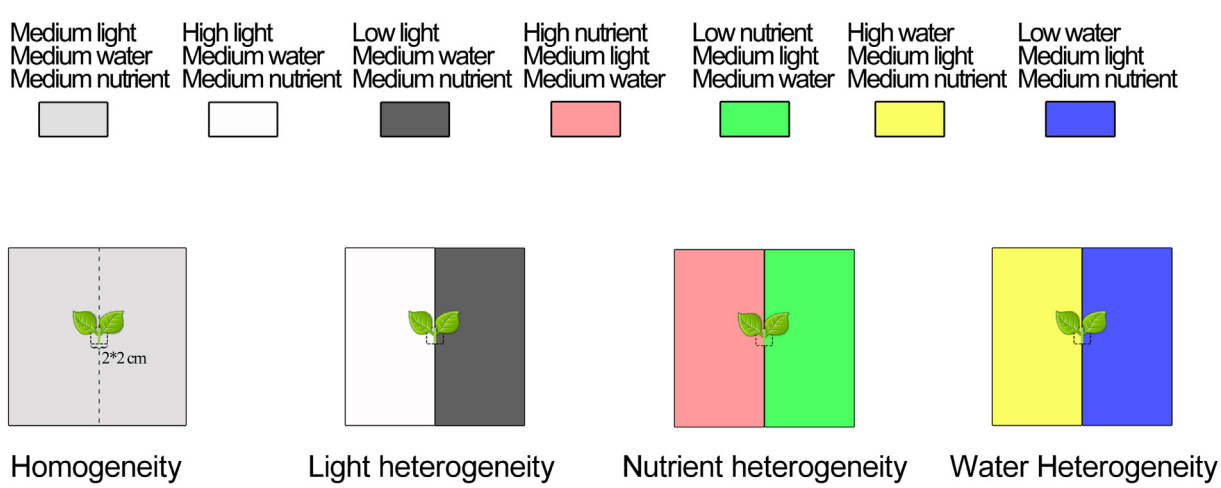

Fig. 1. Experimental design. We started three experiments in parallel, testing effects of origin (invasive alien vs. native) and resource heterogeneity (homogeneity vs. heterogeneity, i.e. homogeneity vs. light heterogeneity, homogeneity vs. nutrient heterogeneity and homogeneity vs. water heterogeneity, respectively) in a homogeneous distribution and three heterogeneous distributions of light, soil nutrient or water.

in random four replicates by a Soil Moisture Meter (TZS-II, HEB Biotechnology Co., Xi'an, China). There were eight replicates of each of the eight treatment combinations (origin × resource heterogeneity) of the experiments, resulting in a total of 32 ramet pairs (pots) for each species and 320 ramets for all ten species.

The experiments were finished after 10 weeks on 12 November 2014. During the experiment, the mean temperature in the glasshouse was 23.9 °C and the relative humidity 75.5% (measured by Amprobe TR300, Amprobe, Everett, WA, USA). Light intensity in the glasshouse was about 70% of that outside.

2.3. Data analysis

We first calculated the number of ramets and harvested biomass from each pot in two parts: the two patches (resource-high and resource-low) of pot (Fig. 1). Both patches of the plant was separated into leaves, roots and clonal growth organs, i.e. stolons for stoloniferous plants or rhizomes for rhizomatous plants in the four treatments. They were dried in an oven at 80 °C for 72 h, and then weighed to obtain dry biomass.

We analyzed this study as three independent experiments as we focused on the differences between homogeneity and heterogeneity of resources (i.e. homogeneity vs. light heterogeneity, homogeneity vs. nutrient heterogeneity and homogeneity vs. light heterogeneity, respectively), but were not interested in testing for differences between the effect of the three resources (light, soil nutrients and water). For each of the three experiments, data (total mass, number of ramet, root mass, clonal organ mass and leaf mass) of plants were analyzed separately with linear mixed models using the R package "nlme" (Pinheiro et al., 2015). In these models, at the whole clone level (i.e. the whole pot), we included species origin in China (invasive alien vs. native), resource heterogeneity (homogeneity vs. heterogeneity) and their two-way interaction as fixed terms. At the patch level, we included the data (total mass, shoot mass and root mass per ramet) on evaluating average performance of single ramet, and we also included species origin, resource heterogeneity, patch quality (resource-high vs. resource-low patches) and their interaction as fixed terms. To account for phylogenetic non-independence of species and for non-independence of the same taxonomic species pair and the replicates of the same species, we included species identity nested within taxonomic pair as random terms in all models. Log-likelihood-ratio was used to evaluate the significance of the fixed terms in our linear mixed models (Zuur et al., 2009). Therefore, a model with the term is comparison with another model without the term, i.e., three-way interaction was exhibited in model where we had removed three-way interaction (origin × heterogeneity × quality) to the full model. Then the calculated log-likelihood ratios are approximately Chi squared (χ^2) distributed (Zuur et al., 2009). As species within each of five pairs of different origin might vary in their responses to the nutrient treatments, we also ran models with random slopes for taxonomic pair and species identity with respect to resource heterogeneity and patch quality. We only conducted the random slopes models for the total mass, because this did not decrease the AIC-values (i.e. increased model fit) for the other traits model (Zuur et al., 2009). All analyses were conducted using the free software R (version 3.0.2; R Development Core Team, 2012).

3. Result

3.1. Plant performance

Overall, invasive species produced significantly more total mass, number of ramets (except for light heterogeneity), root mass, clonal growth organ mass and leaf mass than the native species in both homogeneous and heterogeneous environments at both the patches (Tables 2–4; Fig. 2) and the whole clone level (Table A.1; Fig. 3). Across both origins, nutrient heterogeneity significantly increased total mass and root mass, and marginally increased clonal organ mass and leaf mass as compared to homogeneous treatment (Tables 2–4, Table A.3; Figs. 2, 3).

At the patch level, all biomass parameters and number of ramets significantly increased in resource-high patches in all resource heterogeneous environments, as indicated by significant resource heterogeneity × quality effects (Tables 2–4; Fig. 2). Interestingly, the positive effects of resource-high patches on total mass and number of ramets in three heterogeneous environments, leaf mass in light and nutrient heterogeneity and root mass and clonal organ mass in water heterogeneity were always significantly stronger in invasive alien species than in native species, as indicated by significant origin × resource heterogeneity × quality effects (Tables 2–4; Fig. 2).

The positive effects of nutrient heterogeneity on all biomass parameters of the whole clone were always significantly stronger in invasive alien species than in native species, as indicated by significant origin × nutrient heterogeneity effects (Table A.1, Fig. 3). Root mass in light heterogeneity, and marginally number of ramets in light and water

Table 2
Results of linear mixed models for effects of origin (invasive vs. native), light heterogeneity (homogeneity vs. heterogeneity) and patch quality (resource-high vs. resource-low patches) on total mass, number of ramets, root mass, clonal organ mass and leaf mass at the patch level.

Effect	DF	Total mass		Number of ramets		Root mass		Clonal organ mass		Leaf mass	
		χ^2	*P*	χ^2	*P*	χ^2	*P*	χ^2	*P*	χ^2	*P*
Fixed factor											
Origin (O)	1	**8.58**	**0.003**	2.50	0.114	**4.73**	**0.030**	**6.73**	**0.010**	**6.06**	**0.014**
Light (L)	1	0.48	0.487	1.70	0.192	1.61	0.205	0.10	0.757	0.70	0.403
Quality (Q)	1	**11.24**	**<0.001**	**34.16**	**<0.001**	**10.71**	**0.001**	**6.95**	**0.008**	**8.79**	**0.003**
O × L	1	0.06	0.813	*3.31*	*0.069*	**5.96**	**0.015**	1.33	0.249	0.01	0.916
O × Q	1	2.37	0.124	**6.61**	**0.010**	0.85	0.356	0.46	0.499	*3.33*	*0.068*
L × Q	1	**12.52**	**<0.001**	**27.36**	**<0.001**	**11.05**	**<0.001**	**6.18**	**0.013**	**11.20**	**<0.001**
O × L × Q	1	**4.67**	**0.031**	**17.85**	**<0.001**	2.17	0.141	1.02	0.312	**6.22**	**0.013**
Random factor	N	SD		SD		SD		SD		SD	
Taxonomic pair	5	<0.001*		3.257		0.011		0.020		<0.001	
Species identity	10	0.187*		0.675		0.051		0.038		0.151	
Residual		0.291		1.346		0.063		0.091		0.179	
R^2 of the model											
Marginal R^2		0.201		0.124		0.084		0.055		0.072	
Conditional R^2		0.897		0.811		0.601		0.695		0.717	

Note: *the standard deviation (SD) was only given the intercept of taxonomic pair and species identity here (for the random slopes of resource heterogeneity and patch quality, please, see Table A.5). Values are in bold when $P < 0.05$ and in italic when $0.05 < P < 0.1$. The same below.

Table 3
Results of linear mixed models for effects of origin (invasive vs. native), nutrient heterogeneity (homogeneity vs. heterogeneity) and patch quality (resource-high vs. resource-low patches) on total mass, number of ramets, root mass, clonal organ mass and leaf mass at the patch level.

Effect	DF	Total mass		Number of ramets		Root mass		Clonal organ mass		Leaf mass	
		χ^2	*P*	χ^2	*P*	χ^2	*P*	χ^2	*P*	χ^2	*P*
Fixed factor											
Origin (O)	1	**6.32**	**0.012**	**4.19**	**0.041**	**4.68**	**0.031**	**10.37**	**0.001**	**4.70**	**0.030**
Nutrient (N)	1	**4.18**	**0.041**	2.49	0.114	**5.74**	**0.017**	*2.80*	*0.094*	1.84	0.175
Quality (Q)	1	**12.24**	**<0.001**	**44.50**	**<0.001**	**4.35**	**0.037**	**8.30**	**0.004**	**10.00**	**0.002**
O × N	1	**6.56**	**0.010**	**6.56**	**0.010**	**11.23**	**<0.001**	**5.66**	**0.017**	*3.02*	*0.082*
O × Q	1	1.64	0.201	1.74	0.187	0.48	0.487	1.13	0.289	1.79	0.181
N × Q	1	**15.48**	**<0.001**	**45.24**	**<0.001**	**4.92**	**0.027**	**8.78**	**0.003**	**14.01**	**<0.001**
O × N × Q	1	**3.84**	**0.050**	**8.72**	**0.003**	1.57	0.211	1.99	0.159	**4.27**	**0.039**
Random factor	N	SD		SD		SD		SD		SD	
Taxonomic pair	5	<0.001		2.289		<0.001		<0.001		<0.001	
Species identity	10	0.287		0.536		0.081		0.033		0.204	
Residual		0.305		1.237		0.067		0.113		0.187	
R^2 of the model											
Marginal R^2		0.234		0.132		0.095		0.036		0.079	
Conditional R^2		0.931		0.853		0.677		0.783		0.751	

Values are in bold when $P < 0.05$ and in italic when $0.05 < P < 0.1$.

heterogeneity showed the same patterns to biomass parameters in nutrient heterogeneity (Table A.1, Fig. 3).

3.2. Ramet performance and allocation

Invasive species significantly or marginally accumulated higher total and shoot mass per ramet in light heterogeneity and higher all ramet parameters in nutrient heterogeneity than the native species at both the patches (Tables A.2–A.4; Fig. 4). In resource-low patches, all ramet parameters in nutrient heterogeneity, shoot mass per ramet in light heterogeneity and root mass per ramet in water heterogeneity were significantly higher than those in resource-high patches (interactions of resource heterogeneity × quality effects (Tables A.2–A.4; Fig. 4). Furthermore, the positive effects of resource-low patches on all ramet parameters significantly stronger in invasive alien species than in native species, as indicated by significant origin × resource heterogeneity × quality effects in nutrient heterogeneity (Table A.3; Fig. 4). However, the effects were significant on shoot allocation in the light heterogeneity (Table A.2; Fig. 4) and significant on root allocation or marginal on the whole ramet in water heterogeneity (Table A.4; Fig. 4). Water heterogeneity significantly promoted ramet allocation as compared to homogeneous treatment (Table A.4, Fig. 4).

4. Discussion

Our results showed that plant performance, allocation and selective placement patterns of ramets in response to light, soil nutrient and water heterogeneity differed considerably between the alien invasive and native clonal plants. All biomass parameters and number of ramets significantly increased in resource-high patches in all resource heterogeneous environments. Remarkably, the benefits of invasive alien plants from resource-high patches were significantly higher than native clonal plants in all three types of heterogeneous environments.

Small-scale spatial heterogeneity in the distribution of resource (i.e. light, water and nutrient) is common in natural habitats (Gascha et al., 2015; Jackson and Caldwell, 1993; Peipoch et al., 2016), and plants have developed adaptive plasticity to resource heterogeneity (Alpert, 1999; Chen et al., 2017; De Kroon et al., 1998; He et al., 2011; Roiloa and Retuerto, 2007; Stuefer et al., 1994). Many studies have shown that clonal plants can benefit from resource heterogeneity, especially in resource-rich patches, mediated by clonal plasticity such as clonal integration, selective placement of ramets (De Kroon and Hutchings, 1995; De Kroon et al., 1998; Dong et al., 2015; Van Kleunen and Fischer, 2001). In our study, plants indicated obvious root-foraging and selective placement of ramets (i.e. producing more ramets or higher biomass in the high-quality patches than the low-quality ones) in

Table 4
Results of linear mixed models for effects of origin (invasive vs. native), water heterogeneity (homogeneity vs. heterogeneity) and patch quality (resource-high vs. resource-low patches) on total mass, number of ramets, root mass, clonal organ mass and leaf mass at the patch level.

Effect	DF	Total mass		Number of ramets		Root mass		Clonal organ mass		Leaf mass	
		χ^2	*P*	χ^2	*P*	χ^2	*P*	χ^2	*P*	χ^2	*P*
Fixed factor											
Origin (O)	1	**5.82**	**0.016**	**4.05**	**0.044**	**3.95**	**0.047**	**6.16**	**0.013**	**4.20**	**0.040**
Water (W)	1	0.03	0.874	2.55	0.110	0.08	0.782	1.56	0.212	0.10	0.751
Quality (Q)	1	**10.56**	**0.001**	**34.69**	**<0.001**	**9.55**	**0.002**	**14.10**	**<0.001**	**4.75**	**0.029**
O × W	1	0.01	0.918	*2.93*	*0.087*	0.51	0.476	1.78	0.182	0.40	0.528
O × Q	1	1.2	0.291	*3.39*	*0.065*	0.73	0.392	2.64	0.105	0.29	0.594
W × Q	1	**14.41**	**<0.001**	**36.30**	**<0.001**	**11.81**	**<0.001**	**17.04**	**<0.001**	**7.73**	**0.005**
O × W × Q	1	**4.26**	**0.039**	**16.22**	**<0.001**	*2.94*	*0.086*	**6.01**	**0.014**	2.13	0.144
Random factor	N	SD		SD		SD		SD		SD	
Taxonomic pair	5	<0.001		2.401		<0.001		<0.001		<0.001	
Species identity	10	0.252		0.299		0.068		0.045		0.186	
Residual		0.281		1.613		0.066		0.090		0.170	
R2 of the model											
Marginal R^2		0.269		0.164		0.053		0.047		0.070	
Conditional R^2		0.912		0.738		0.686		0.633		0.799	

Values are in bold when $P < 0.05$ and in italic when $0.05 < P < 0.1$.

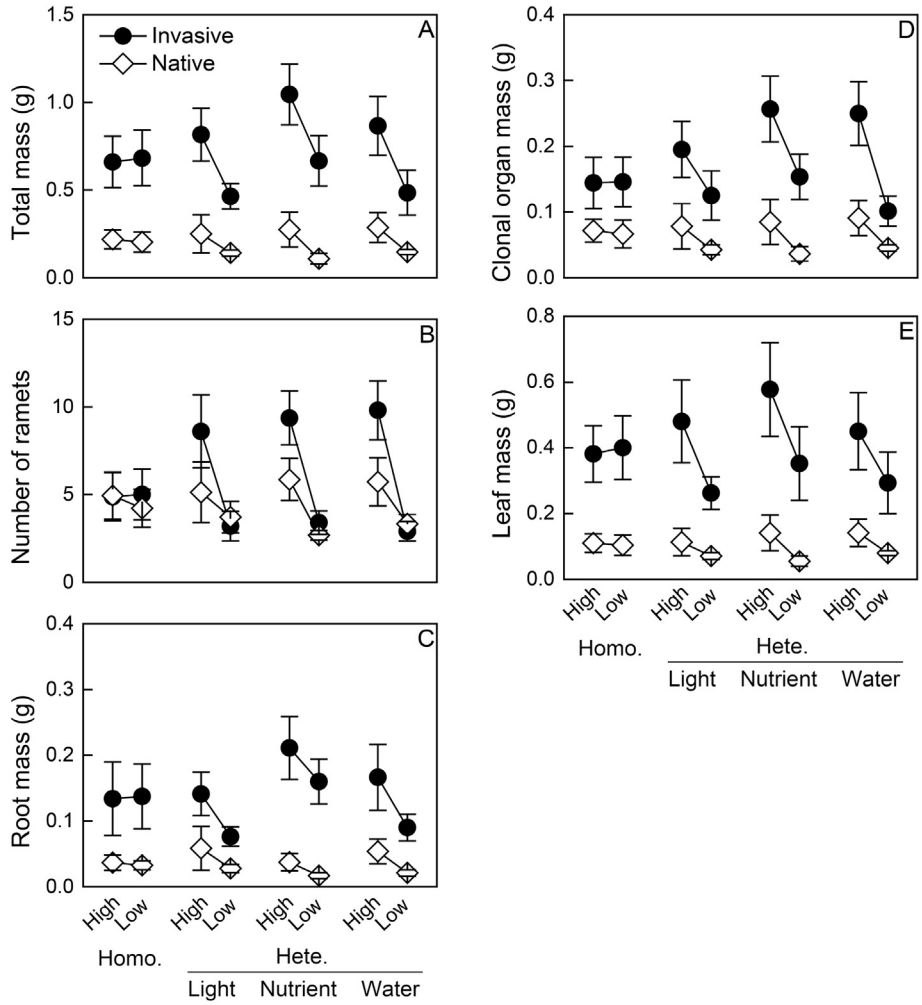

Fig. 2. Total mass (A), number of ramet (B), root mass (C), clonal organ mass (D) and leaf mass (E) of invasive and native plant species in the resource-high or resource-low patches in the homogeneity, and light, nutrient or water heterogeneity.

heterogeneous light, soil water and nutrient environments. Nutrient heterogeneity significantly promotes overall biomass, root mass and number of ramets of the whole plant compared with that in homogeneous treatment. Previous studies showed that preferential selective placements of ramets and root in nutrient-rich patches promoted the amount and efficiency of resource capture by these ramets and further increased local growth (De Kroon and Hutchings, 1995; Dong et al., 2015; Van Kleunen and Fischer, 2001). Furthermore, their efficiency of resource capture in nutrient-high patches can also benefit growth of the whole plant, partly due to resource sharing between ramet systems (He et al., 2011; Song et al., 2013; Zhou et al., 2012). However, in light and water heterogeneity, number and biomass of ramets were higher in resource-high patches and lower in resource-low patches compared with that in the homogeneous treatment. Consequently, there was no significant difference in the growth of the whole plants between the homogeneous and the heterogeneous light or water treatments.

Remarkably, we found that alien invasive clonal plants benefited more in biomass and ramet productions from high-quality patches than native ones in heterogeneous light, soil nutrient and water environments, and invasive plants benefit more from nutrient heterogeneity than native in the whole plant. This result suggests that in environments with heterogeneous distribution of light, soil water and nutrients (commonly encountered by alien plants during their invasion into a new range), root-forging and consequently preferential selective placement of ramets after new ramets system production confers alien clonal plants with growth advantages over native ones being (Burns, 2008; Dawson et al., 2012; Melbourne et al., 2007; Parepa et al., 2013; Pearse and Altermatt, 2013). Some studies have reported a stronger positive effect of resource heterogeneity on growth performance in invasive alien than in native species (Wang et al., 2017; You et al., 2014), and a stronger root forging and selective placement of ramets benefit growth performance in invasive alien clonal plants in resource heterogeneity (Dong et al., 2015; Keser et al., 2014, 2015). We also found that invasive alien plants produced more biomass than native plants, which is well in line with previous findings that fast biomass production and resource uptake provides the alien species with the growth advantage compared with native species in communities (Van Kleunen et al., 2010; Wang et al., 2017). This is also well consistent with the evolution of increased competitive ability (EICA) hypothesis that decreased defense due to the absence of specialist herbivores and increased investment to growth and reproduction (Blossey and Nötzold, 1995; Richardson et al., 2000). Our study supports the idea that the potential for strong clonal plasticity performance plays a role in plant invasions (Hulme, 2008; Keser et al., 2014; Richards et al., 2006; Van Kleunen et al., 2000).

Overall, invasive plants had higher biomass parameters per ramet than natives. Furthermore, ramet parameters of invasive plants benefited from resource-low patches were significantly higher than natives. Three different selective placement patterns of ramets in resource-low patches were exhibited in invasive clonal plants as consequences of different resource heterogeneity: the complete pattern that single ramet investment increased in nutrient heterogeneity; the

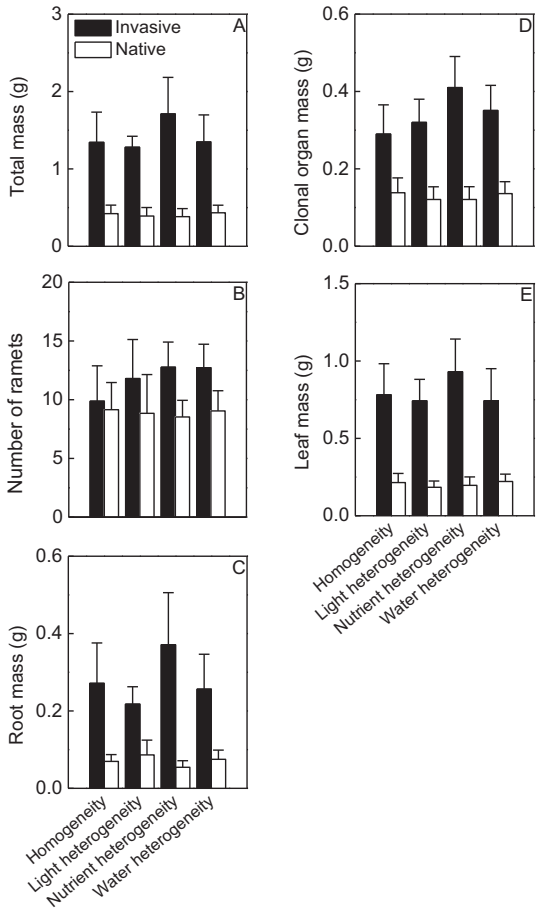

Fig. 3. Total mass (A), number of ramet (B), root mass (C), clonal organ mass (D) and leaf mass (E) of invasive and native plant species in the homogeneity, and light, nutrient or water heterogeneity.

above pattern that ramet shoot investment increased in light heterogeneity; and the below pattern ramet root investment increased in water heterogeneity. Studies indicated low-resource environment promoted growth superior of invasive plants over natives due to the higher resource-use efficiency (Funk and Vitousek, 2007). It suggested that species with the capacity to maintain high growth performance in stressful or resource-poor environments should be more likely to be competitive advantages in heterogeneity environments (Richards et al., 2006; Funk and Vitousek, 2007; Van Kleunen and Fischer, 2001). In this study, investment on less, larger ramet was the adaptive strategy of invasive plants in resource-poor patches. Pattern of less and larger ramet would be higher efficiency in resource capture than that of more and small ramets in resource-poor habitats. Pattern of less and larger ramet also had high plasticity in shoot-root allocation to coping with light, water or nutrient heterogeneity. The patterns were also consistent with the available resources in space, i.e., light-shoot related, and water-root related, which indicated the higher resource-use efficiency in resource heterogeneity environments.

5. Conclusion

Our results indicated that not ramet but the whole plant of invasive alien plants benefited higher from high and/or low resource patches than native clonal plants in all three types of heterogeneous environments. Different placement patterns of ramets in resource-low patches in invasive plants that higher benefits of below- and above-ground of ramet in nutrient heterogeneity, of aboveground in light heterogeneity and of belowground in water heterogeneity, which suggest that adaptively selective placement patterns of ramets promoted a higher

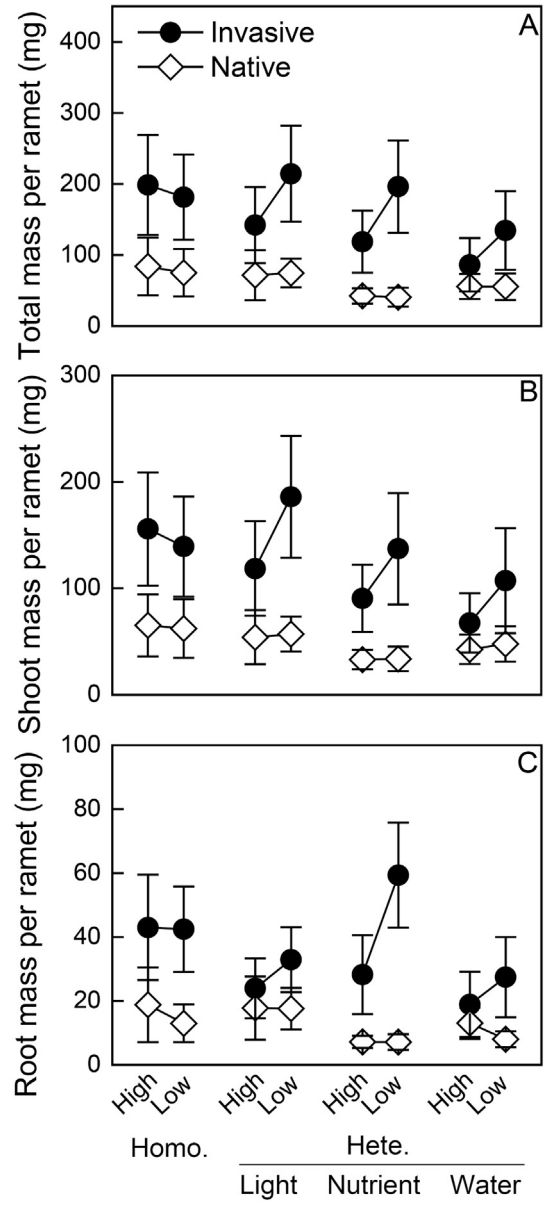

Fig. 4. Total mass per ramet (A), shoot mass per ramet (B) and root mass per ramet (C) of invasive and native plant species in the resource-high or resource-low patches in the homogeneity, and light, nutrient or water heterogeneity.

morphology plasticity and performance in invasive clonal plants over natives. When alien clonal plants spread new areas with light, soil nutrients or water heterogeneity, selective placement patterns of ramets might play an important role in competitive superior by capitalizing more on additional resources.

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Author contributions

Y.J.W. and F.H.Y. designed the experiment and did the statistical analysis, X.H.Y., A.M.C., D.C. and C.G.L. performed the

experiment, D.C., A.A. and Y.J.W. wrote the first draft of the manuscript, F.H.Y. and Z.X.Z. contributed substantially to the revisions.

Competing financial interests

The authors declare no competing financial interests.

Appendix A

Table A.1
Results of generalized linear mixed models for effects of origin (invasive vs. native) and resource heterogeneity (homogeneity vs. heterogeneity) on total mass, number of ramets, root mass, clonal organ mass and leaf mass of the whole clone in (A) light, (B) nutrient and (C) water experiments.

Effect	DF	Total mass		Number of ramets		Root mass		Clonal organ mass		Leaf mass	
		χ^2	*P*	χ^2	*P*	χ^2	*P*	χ^2	*P*	χ^2	*P*
(A)											
Fixed factor											
Origin (O)	1	**8.580**	**0.003**	2.50	0.114	**4.727**	**0.030**	**6.730**	**0.010**	**6.062**	**0.014**
Light heterogeneity (L)	1	0.489	0.484	1.66	0.198	1.571	0.210	0.082	0.775	0.842	0.359
O × L	1	0.057	0.812	*3.24*	*0.072*	**5.890**	**0.015**	1.139	0.286	0.013	0.908
Random factor	N	SD		SD		SD		SD		SD	
Taxonomic pair	5	<0.001		6.514		0.022		0.041		<0.001	
Species identity	10	0.373		1.289		0.102		0.074		0.302	
Residual		0.291		1.346		0.063		0.091		0.179	
(B)											
Fixed factor											
Origin (O)	1	**6.319**	**0.012**	**4.19**	**0.041**	**4.675**	**0.031**	**10.367**	**0.001**	**4.699**	**0.030**
Nutrient heterogeneity (N)	1	**4.670**	**0.031**	2.619	0.106	**5.742**	**0.017**	*2.904*	*0.088*	*2.785*	*0.095*
O × H	1	**7.501**	**0.006**	**7.04**	**0.008**	**11.598**	**<0.001**	**5.952**	**0.015**	**4.643**	**0.031**
Random factor	N	SD		SD		SD		SD		SD	
Taxonomic pair	5	<0.001		4.578		<0.001		<0.001		<0.001	
Species identity	10	0.576		1.023		0.162		0.067		0.410	
Residual		0.305		1.237		0.067		0.113		0.187	
(C)											
Fixed factor											
Origin (O)	1	**5.821**	**0.016**	**4.05**	**0.044**	**3.950**	**0.047**	**6.157**	**0.013**	**4.200**	**0.040**
Water heterogeneity (W)	1	0.025	0.874	2.62	0.105	0.083	0.774	1.466	0.226	0.124	0.725
O × W	1	0.010	0.919	*3.05*	*0.081*	0.545	0.460	1.688	0.194	0.484	0.487
Random factor	N	SD		SD		SD		SD		SD	
Taxonomic pair	5	<0.001		4.801		<0.001		<0.001		<0.001	
Species identity	10	0.503		0.455		0.136		0.090		0.372	
Residual		0.281		1.613		0.066		0.089		0.170	

Values are in bold when $P < 0.05$ and in italic when $0.05 < P < 0.1$.

Table A.2
Results of linear mixed models for effects of origin (invasive vs. native), light heterogeneity (homogeneity vs. heterogeneity) and patch quality (resource-high vs. resource-low patches) on mass per ramet, shoot per ramet, and root per ramet at the patch level.

Effect	DF	Total mass per ramet		Shoot mass per ramet		Root mass per ramet	
		χ^2	*P*	χ^2	*P*	χ^2	*P*
Fixed factor							
Origin (O)	1	*3.55*	*0.060*	**3.98**	**0.046**	2.00	0.157
Light (L)	1	0.48	0.487	0.02	0.876	**4.33**	**0.038**
Quality (Q)	1	0.88	0.349	1.36	0.244	0.04	0.835
O × L	1	0.04	0.835	0.35	0.555	**7.33**	**0.007**
O × Q	1	1.37	0.242	1.37	0.241	1.49	0.223
L × Q	1	*3.81*	*0.051*	**4.35**	**0.037**	1.66	0.198
O × L × Q	1	2.27	0.132	**3.80**	**0.049**	0.11	0.741
Random factor	N	SD		SD		SD	
Taxonomic pair	5	<0.001		<0.001		<0.001	
Species identity	10	0.080		0.062		0.019	
Residual		0.069		0.057		0.013	

Values are in bold when $P < 0.05$ and in italic when $0.05 < P < 0.1$.

Table A.3
Results of linear mixed models for effects of origin (invasive vs. native), nutrient heterogeneity (homogeneity vs. heterogeneity) and patch quality (resource-high vs. resource-low patches) on mass per ramet, shoot per ramet, and root per ramet at the patch level.

Effect	DF	Total mass per ramet		Shoot mass per ramet		Root mass per ramet	
		χ^2	*P*	χ^2	*P*	χ^2	*P*
Fixed factor							
Origin (O)	1	*2.90*	*0.089*	*2.76*	*0.097*	*3.28*	*0.07*
Nutrient (N)	1	2.17	0.141	**3.84**	**0.050**	0.03	0.860
Quality (Q)	1	1.32	0.250	0.62	0.432	*2.98*	*0.084*
O × N	1	0.84	0.360	0.16	0.688	*3.82*	*0.051*

(continued on next page)

Table A.3 (*continued*)

Effect	DF	Total mass per ramet		Shoot mass per ramet		Root mass per ramet	
		χ^2	*P*	χ^2	*P*	χ^2	*P*
O × Q	1	*2.71*	*0.099*	0.86	0.353	**6.67**	**0.010**
N × Q	1	**6.01**	**0.014**	**3.86**	**0.049**	**7.73**	**0.005**
O × N × Q	1	**4.68**	**0.031**	*3.14*	*0.076*	**4.08**	**0.043**
Random factor	N	SD		SD		SD	
Taxonomic pair	5	<0.001		<0.001		<0.001	
Species identity	10	0.103		0.076		0.026	
Residual		0.099		0.071		0.018	

Values are in bold when $P < 0.05$ and in italic when $0.05 < P < 0.1$.

Table A.4
Results of linear mixed models for effects of origin (invasive vs. native), water heterogeneity (homogeneity vs. heterogeneity) and patch quality (resource-high vs. resource-low patches) on mass per ramet, shoot per ramet, and root per ramet at the patch level.

Effect	DF	Total mass per ramet		Shoot mass per ramet		Root mass per ramet	
		χ^2	*P*	χ^2	*P*	χ^2	*P*
Fixed factor							
Origin (O)	1	2.43	0.119	2.20	0.138	2.45	0.118
Water (W)	1	**10.15**	**0.001**	**10.72**	**0.001**	**8.02**	**0.005**
Quality (Q)	1	0.29	0.592	0.07	0.789	0.54	0.463
O × W	1	1.92	0.165	*2.93*	*0.087*	1.30	0.254
O × Q	1	1.00	0.317	*3.18*	*0.075*	0.43	0.510
W × Q	1	*3.74*	*0.053*	1.05	0.305	**4.04**	**0.044**
O × W × Q	1	*2.706*	*0.100*	1.02	0.313	**3.77**	**0.049**
Random factor	N	SD		SD		SD	
Taxonomic pair	5	<0.001		<0.001		<0.001	
Species identity	10	0.091		0.022		0.069	
Residual		0.082		0.017		0.060	

Values are in bold when $P < 0.05$ and in italic when $0.05 < P < 0.1$.

Table A.5
The standard deviation (SD) for random effects of the total mass model, which included random slopes for taxonomic pair and species identity with respect to resource heterogeneity (i.e. light heterogeneity, nutrient heterogeneity and water heterogeneity) and patch quality (resource-high and resource-low patches).

Treatment	Standard deviation (SD)	
	Taxonomic pair	Species identity with taxonomic pair
Intercept (i.e. homogeneity)	0.369	0.513
Light heterogeneity	0.118	0.675
Nutrient heterogeneity	0.177	0.732
Water heterogeneity	0.154	0.701
Intercept (i.e. resource-low)	0.193	0.622
Resource-high in light heterogeneity	0.257	0.583
Intercept (i.e. resource-low)	0.202	0.614
Resource-high in nutrient heterogeneity	0.221	0.655
Intercept (i.e. resource-low)	0.160	0.727
Resource-high in water heterogeneity	0.206	0.683

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